



Consolidated NLP Practice Questions for B.Tech CSE (Midterm Topics)

This list combines all distinct questions from the provided exam papers and important question lists, categorized by expected weight (2 Marks and 5 Marks).

Part A: Short Answer Questions (Approx. 2 Marks)

(Focus on definitions, brief explanations, or short lists with examples.)

1. Core Concepts & Analysis

1. Define tokenization.
2. Write about word tokenization.
3. Define morpheme. Write about types of morphemes with example.
4. What is irregularity? Impact of Irregularity on NLP.
5. What is Ambiguity? Write about types of Ambiguity.
6. Write about pragmatic analysis and its key tasks.
7. Write about NLTK.

2. Preprocessing & Parsing

8. Write about n-gram tokenization with example.
9. Write about four NLP text preprocessing techniques.
10. Write about Porter's Stemmer.
11. Shortly differentiate between deep and shallow parsing.
12. What is a tree bank?

3. Language Modeling (LM)

13. Define language modelling.
14. What is the n-gram model? (or n-gram language models).
15. Write about challenges in Language Modelling.
16. What do you mean by parameter estimation?
17. Briefly define coverage rate and perplexity.
18. Define language model evaluation.

4. Semantic Analysis

19. Write about types of semantic parsing techniques with advantage and disadvantages.
20. Write about word sense.
21. What is predicate argument structure and its goal. Why it matters in NLP?
22. Write difference between Deep Semantic Parsing and Shallow Semantic Parsing.
23. Define word knowledge.

24. Why is semantic analysis difficult?
25. What is semantic interpretation?
26. Define Hyponymy and Homonymy.
27. Write about system paradigms.
28. Write about some examples for early NLP systems.
29. Write about Bayesian Parameter Estimation.

Part B: Essay Questions (Approx. 5 Marks)

(Focus on detailed explanations, comparisons, process descriptions, or problem-solving.)

1. Core NLP and Applications

30. Explain about **various phases of NLP**.
31. Write about **applications of NLP**.
32. Write about **NLP pre-processing techniques** in detail with libraries used.
33. Why is **document structure important in NLP**? Discuss the different methods used to identify the structure of a document.

2. Morphology & Lexical Processing

34. Explain the difference between **stemming and lemmatization** with examples.
35. Explain **various stemming and lemmatization techniques**.
36. Explain the **Dictionary Lookup model** for morphological analysis in NLP. How can a custom dictionary be created? Illustrate with an example.
37. Explain about various **Morphological Models**.

3. Parsing and Syntax

38. Explain **Top-Down and Bottom-Up parsing techniques** with examples.
39. Compare **constituency parsing with dependency parsing methods** with examples.
40. Describe **shift-reduce parsing techniques**.
 - **Practice Problem 1 (From Mid 1):** Perform Shift Reduce parsing for the input string "id+id+id" with the Grammar: $E \rightarrow E+T \mid T, T \rightarrow T * F \mid F, F \rightarrow \text{text{id}}$.
 - **Practice Problem 2 (From Mid 1 Retest):** Perform Shift Reduce parsing for input string "id + id + id" with the Grammar: $S \rightarrow S + S, S \rightarrow S * S, S \rightarrow \text{text{id}}$.
41. **CYK parsing (chart parsing)**.
42. What is **Sentence ambiguity parsing**? Explain various ambiguity resolution models.
43. What are the **Limitations in syntax parsing**?
44. Write about **Probabilistic Context Free Grammar (PCFG) model** for Resolutions of ambiguity parsing. (Practice with example problems).
 - **Practice Problem 3 (From Mid 2):** Use the provided PCFG grammar to parse the target sentence: $\text{"I saw the man with the telescope"}$.

4. Advanced Language Modeling

- 45. Write about **language modelling**.
- 46. Explain **three types of language models**.
- 47. Explain **class based language model**.
- 48. Write about **multi lingual and cross lingual language modelling**.
- 49. **Bi-gram Model Problem:** Implement Bi-gram model to calculate the probability of the given test sentence.
 - **Training Set:** She is pretty; she is not a robot; she lives in Hyderabad.
 - **Test Sentence:** she is not.

5. Semantic Analysis and WSD

- 50. Explain about **Word Sense Disambiguation (WSD)** and various approaches to WSD.
- 51. Explain **Extended Lesk Algorithm** with example evaluation.
- 52. Explain **Lesk Algorithm** with example evaluation and mention advantages and disadvantages.
- 53. Write about a **semantic similarity based approach for WSD**.
- 54. Explain difference between **FrameNet and PropBank**.
- 55. Write about **FrameNet and PropBank**.
- 56. Explain about **predicate argument structure**.
- 57. Explain in detail about **classification paradigm of semantic argument**.
- 58. Write about **Meaning Representation Systems** and its types in detail.
- 59. Explain in detail about **meaning representation in semantic parsing**.
- 60. What are the **types of representations of semantic parsing**?
- 61. What **requirements are necessary for achieving semantic representation**?
- 62. Provide a detailed description of **meaning representation and its need**.
- 63. Explain **system architectures in semantic parsing**.